1 Interpreting the Wasserstein Decomposition

This section illustrates how to interpret the decomposition of the integrated Wasserstein treatment effect. For a scalar outcome Y, binary treatment A, and covariates X, define

$$\Psi^2 = \int_{\mathcal{X}} \int_0^1 \{Q_1(u \mid x) - Q_0(u \mid x)\}^2 \, du \, dP_X(x),$$

where $Q_a(u \mid x)$ is the u-th conditional quantile of Y given $A = a$ and $X = x$. Let

$$\Delta(x, u) = Q_1(u \mid x) - Q_0(u \mid x)$$

denote the conditional quantile treatment effect.

The decomposition separates Ψ^2 into four interpretable components:

$$\Psi^2 = \tau^2 + \mathbb{V}\{\tau(X)\} + \theta_b + \theta_c.$$

Here,

$$\tau(x) = \int_0^1 \Delta(x, u) \, du$$

is the quantile-averaged treatment effect at covariate value x, and

$$\tau = \int_{\mathcal{X}} \tau(x) \, dP_X(x)$$

is the overall average additive treatment effect. The term τ^2 measures the squared average additive treatment effect (across all covaraiates and quantiles), while

$$\mathbb{V}\{\tau(X)\}$$

measures heterogeneity in the quantile-averaged treatment effect across covariate values.

Next, define

$$\overline{\Delta}(u) = \int_{\mathcal{X}} \Delta(x, u) \, dP_X(x).$$

The term

$$\theta_b = \mathbb{V}\{\overline{\Delta}(U)\}$$

measures rank or quantile-index heterogeneity. If θ_b is large, then the average treatment effect differs across the outcome distribution. For example, treatment may have little effect near the median but larger effects in the tails.

Finally,

$$\theta_c = \mathbb{E}\left[\left\{\Delta(X, U) - \tau(X) - \overline{\Delta}(U) + \tau\right\}^2\right]$$

measures covariate–quantile interaction. If θ_c is large, then the conditional quantile treatment effect cannot be represented solely by an overall additive effect, a covariate-specific additive effect, and a quantile-specific effect. In other words,

$$\Delta(x, u) \neq \tau(x) + \overline{\Delta}(u) - \tau.$$

This indicates that the shape of the treatment effect across quantiles depends on the covariate value.

1.1 Examples

The following examples use simple normal distributions to show how each component of the decomposition behaves. In all three examples,

$$X \sim \text{Uniform}\{1,2,3\}, \qquad A \in \{0,1\},$$

and the untreated conditional distributions are

$$Y \mid X=1, A=0 \sim N(20, 1^2),$$

$$Y \mid X=2, A=0 \sim N(30, 1^2),$$

$$Y \mid X=3, A=0 \sim N(40, 1^2).$$

For a normal conditional distribution,

$$Q_a(u \mid x) = \mu_a(x) + \sigma_a(x)\Phi^{-1}(u),$$

so the conditional quantile treatment effect can be written as

$$\Delta(x,u) = \{\mu_1(x) - \mu_0(x)\} + \{\sigma_1(x) - \sigma_0(x)\}\Phi^{-1}(u).$$

This representation allows us to design examples where different components of the decomposition dominate.

1.1.1 Example 1: Homogeneous Additive Effect

First suppose treatment shifts every conditional distribution upward by 10 without changing its variance:

$$Y \mid X=1, A=1 \sim N(30, 1^2),$$

$$Y \mid X=2, A=1 \sim N(40, 1^2),$$

$$Y \mid X=3, A=1 \sim N(50, 1^2).$$

Then

$$\Delta(x,u) = 10$$

for every x and u. Therefore,

$$\tau = 10, \qquad \tau^2 = 100, \qquad \mathbb{V}\{\tau(X)\} = 0, \qquad \theta_b = 0, \qquad \theta_c = 0.$$

Thus,

$$\Psi^2 = \tau^2 = 100.$$

This is the simplest case: the Wasserstein effect is entirely explained by a constant additive treatment effect.

1.1.2 Example 2: Additive Covariate and Rank Heterogeneity

Next suppose treatment changes the conditional mean differently across covariate values and also increases the variance by the same amount for every covariate value:

$$Y \mid X = 1, A = 1 \sim N(16, 4^2),$$

$$Y \mid X = 2, A = 1 \sim N(30, 4^2),$$

$$Y \mid X = 3, A = 1 \sim N(44, 4^2).$$

The mean shifts are

$$k_1 = -4, \qquad k_2 = 0, \qquad k_3 = 4,$$

and the standard deviation changes are constant:

$$d_1 = d_2 = d_3 = 3.$$

Therefore,

$$\Delta(x, u) = k_x + 3\Phi^{-1}(u).$$

The quantile-averaged treatment effect varies across covariates because k_x depends on x, so

$$\mathbb{V}\{\tau(X)\} > 0.$$

The treatment also changes the spread of the outcome distribution. Since the standard deviation change is the same for all covariate values, this produces rank heterogeneity but no covariate–quantile interaction. In this example,

$$\tau = 0, \qquad \tau^2 = 0,$$

$$\mathbb{V}\{\tau(X)\} = \frac{(-4)^2 + 0^2 + 4^2}{3} = \frac{32}{3},$$

$$\theta_b = 3^2 = 9, \qquad \theta_c = 0.$$

Thus,

$$\Psi^2 = 0 + \frac{32}{3} + 9 + 0 = \frac{59}{3}.$$

This example is additively separable: the treatment effect can be written as a covariate-specific effect plus a quantile-specific effect.

1.1.3 Example 3: Additive Effects and Covariate–Quantile Interaction

Finally, suppose treatment keeps the same covariate-specific mean shifts as in Example 2, but the variance change now depends on X:

$$Y \mid X = 1, A = 1 \sim N(16, 1^2),$$

$$Y \mid X = 2, A = 1 \sim N(30, 7^2),$$

$$Y \mid X = 3, A = 1 \sim N(44, 1^2).$$

The mean shifts are again

$$k_1 = -4, \qquad k_2 = 0, \qquad k_3 = 4.$$

However, the standard deviation changes are now

$$d_1 = 0, \qquad d_2 = 6, \qquad d_3 = 0.$$

Therefore,

$$\Delta(x, u) = k_x + d_x \Phi^{-1}(u).$$

The quantile-averaged treatment effect is the same as in Example 2, so

$$\mathbb{V}\{\tau(X)\} = \frac{32}{3}.$$

However, the standard deviation change now differs by covariate value. The average standard deviation change is

$$\overline{d} = \frac{0 + 6 + 0}{3} = 2.$$

Thus,

$$\theta_b = \overline{d}^2 = 4,$$

while the covariate–quantile interaction component is

$$\theta_c = \frac{(0-2)^2 + (6-2)^2 + (0-2)^2}{3} = 8.$$

Therefore,

$$\tau = 0, \qquad \tau^2 = 0,$$

and

$$\Psi^2 = 0 + \frac{32}{3} + 4 + 8 = \frac{68}{3}.$$

This example has the same covariate-specific mean shifts as Example 2, but the variance change is concentrated in the $X = 2$ group. As a result, the shape of the treatment effect across quantiles depends on the covariate value, producing a nonzero covariate–quantile interaction term.

REMEMBER TO WRITE PACKAGE CODE FOR EACH EXAMPLE!!

MEETING NOTES: Find good R packages for formatting (look through others) (see Kyle's Githbut link) Make github so you can push code sink horn estimation in R

https://cran.r-project.org/web/packages/transport/index.html https://cran.r-project.org/web/packages/T4transp https://cran.r-project.org/web/packages/approxOT/index.html

Find applied example, rank heterogeneity, interesting L-moments default to trapezoidal integration chernozukov implementation for quantile reordering (check estimation section) set default for slope floor in density estimation function, small

replace logistic regression with super learner ensemble method for nuisance functions, check sl.lib, people specify library, for fit$_{m}eanalso, seeifpeoplecanprovidetheirownlearner$

Allow chat

double check whether sparsity function is used in eta construction

crossfit wasserstein nuisance u value, add trimming for u quantiles, see about specifying truncation parameter instead of grid, u should be specified, specify parameter in fit$_{q}uantile_{g}rid$,

A can be arbitrary factor variable, handles multiple data types

check slope$_{f}loorassignment < -1e^{-}6..., alsocheckpropensitybound.Lowerdefaultparameterto0.00001(orlower).I$

Check get$_{t}au_{x}defaultestimation, shouldn'tbeaveragingoverquantiles.UsedoublyrobustestimatorforCATE, Ken$

Check V(tau(X)) estimate, matches Levy 2021, UPDATE: This matches the paper.

Check estimation of V[c], V[b] terms compared to paper, relabel V[c] (use theta c, theta b), additionally in one step decomposition, UPDATE: The estimates were correct.

change examples after setting default u-grid

check n effect on simulation, edge effects large, use large n or lower grid 0.05 - 0.95

example visuals should use appropriate packages for effect plots, check standard for visuals in R packages

Try to make vignette very readable, don't just copy paste.

Check what tau2 does